

Digital Logic Design

Quiz 04

2025/05/19

姓名:

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1. Show that the characteristic equation for the complement output of a T flip-flop is $Q(t+1) = TQ' + T'Q$ (20%)

5.3 $Q'(t+1) = (TQ' + T'Q)' = (TQ')' \cdot (T'Q)' = (T' + Q)(T + Q') = TQ + T'Q'$

2. A certain flip-flop has four operations: complement, set to 1, set to 0, and no change, when inputs A and B are 00, 01, 10, and 11, respectively. (20%)
 - (a) Tabulate the characteristic table.
 - (b) Tabulate the excitation table.
 - (c) Derive the characteristic equation.
 - (d) Show how the flip-flop can be converted to a T flip-flop.

5.4 (a) Characteristic Table:

A	B	Q(t+1)
0	0	Q'(t)
0	1	1
1	0	0
1	1	Q(t)

(b) Excitation Table

Q(t)	Q(t+1)	A	B
0	0	1	X
0	1	0	X
1	0	X	0
1	1	X	1

(c) Derivation of characteristic equation:

A	B	Q(t)	Q(t+1)
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

		B			
		00	01	11	10
A	0	m_0 1	m_1 0	m_3 1	m_2 1
	1	m_4 0	m_5 0	m_7 1	m_6 0
		Q(t)			

$$Q(t+1) = A'Q' + BQ$$

(d) Connect A and B together and apply complement of T input to it.

3. A sequential circuit has two *JK* flip-flops *A* and *B* and one input *x*. The circuit is described by the following flip-flop input equations:(10%)

$$J_a = x' \quad K_a = B'$$

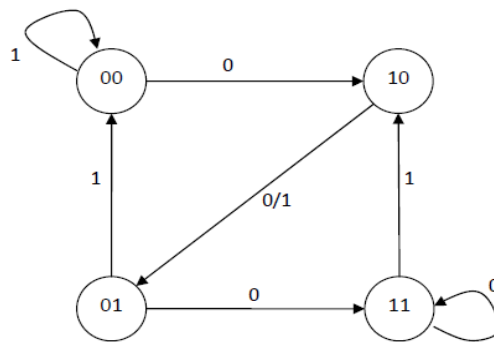
$$J_\beta = A \quad K_\beta = x$$

- (a) Derive the state equations $A(t+1)$ and $B(t+1)$ by substituting the input equations for the *J* and *K* variables.
 (b) Draw the state table and state diagram of the circuit.

5.9 (a) $J_A = x', K_A = B'; A(t+1) = JA' + K'A = x'A' + BA$
 $J_B = A, K_B = x; B(t+1) = JB' + K'B = AB' + x'B$

(b)

Present State		Input	Next state	
<i>A(t)</i>	<i>B(t)</i>	<i>x</i>	<i>A(t+1)</i>	<i>B(t+1)</i>
0	0	0	1	0
0	0	1	0	0
0	1	0	1	1
0	1	1	0	0
1	0	0	0	1
1	0	1	0	1
1	1	0	1	1
1	1	1	1	0



4. Starting from state *a*, and the input sequence 01110010011, determine the output sequence for
 (a) The state table of the previous problem.
 (b) The reduced state table from the previous problem. Show that the same output sequence is obtained for both.(20%)

5.13 Equivalent

(a) State: *a f b c e d g h g g h a*
 Input: 0 1 1 1 0 0 1 0 0 1 1
 Output: 0 1 0 0 0 1 1 1 0 1 0

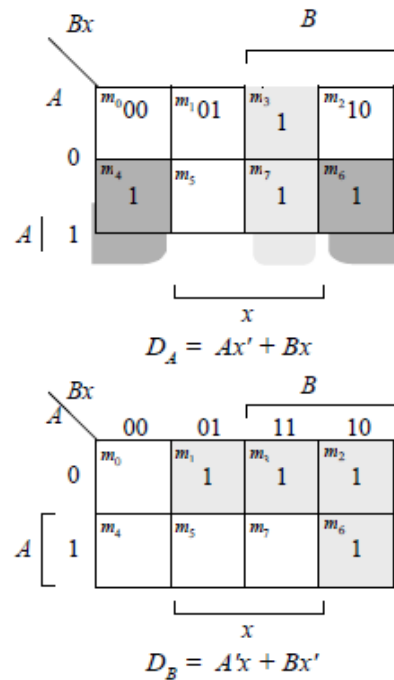
(b) State: *a f b a b d g d g g d a*
 Input: 0 1 1 1 0 0 1 0 0 1 1
 Output: 0 1 0 0 0 1 1 1 0 1 0
 Equivalent states: *b = e, a = c, h = d*

5. Design a sequential circuit with two D flip-flops A and B , and one input x_{in} . (20%)
- (a) When $x_{in} = 0$, the state of the circuit remains the same. When $x_{in} = 1$, the circuit goes through the state transitions from 00 to 01, to 11, to 10, back to 00, and repeats.
- (b) When $x_{in} = 0$, the state of the circuit remains the same. When $x_{in} = 1$, the circuit goes through the state transitions from 00 to 11, to 01, to 10, back to 00, and repeats.

5.16 (a) $D_A = Ax' + Bx$

$$D_B = A'x + Bx'$$

Present state		Input	Next state	
A	B		A	B
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	1
1	0	0	1	0
1	0	1	0	0
1	1	0	1	1
1	1	1	1	0



(b) $D_A = A'x + Ax'$

$$D_B = AB + Bx'$$

Present state		Input	Next state	
A	B		A	B
0	0	0	0	0
0	0	1	1	1
0	1	0	0	1
0	1	1	1	0
1	0	0	1	0
1	0	1	0	0
1	1	0	1	1
1	1	1	0	1

